

XIYUAN (TOMMY) GUO

B.A. Candidate in Computer Science, University of Virginia

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RESEARCH FOCUS

Convex optimization; computational imaging; missing-data imputation; quantum error correction.

EDUCATION

University of Virginia

B.A. in Computer Science

Expected May 2027

Charlottesville, VA

- **GPA:** 3.73/4.00
- **Relevant Coursework:** Graduate Quantum Computing (PHYS 5880, in progress).
- **Independent Study:** Quantum Information (Preskill lecture series); Quantum Mechanics (Griffiths, Sakurai).

Kimball Union Academy

High School Diploma

2024

Meriden, NH

GRANTS AND FELLOWSHIPS

- **Double Hoo Research Grant**, University of Virginia 2026
Secured competitive grant funding to drive independent, interdisciplinary computational research alongside graduate mentorship.

PUBLICATIONS & MANUSCRIPTS IN PREPARATION

- Guo, X., [Co-authors], Ke, L. "Transition-metal magnetic anisotropy and its relationship to structural motifs." *[Manuscript in preparation]*.
- Guo, X., [Co-authors], Liu, M. "Joint Spectro-Polarimetric Lensless Reconstruction via Conic-Constrained ADMM." *[Manuscript in preparation]*. Code available on request.
- Guo, X., [Co-authors], Tong, X.C. "PRISM: Lagrangian Projection Engines for Multivariate Longitudinal Missing Data Imputation." *[Manuscript in preparation]*. Code: github.com/xguot/prism.

PROFESSIONAL RESEARCH EXPERIENCE

Research Assistant

Yale University (Prof. Mengxia Liu's Group)

Feb 2026 – Present

New Haven, CT

- Built a joint spectro-polarimetric forward model recovering a 324-channel datacube (spatial $\times$ 81 wavelengths $\times$ 4 Stokes parameters) from a single multiplexed lensless speckle image.
- Formulated the reconstruction as a conic-constrained inverse problem: enforcing a Lorentz cone constraint on the Stokes vector ($S_0 \geq \|S_{1:3}\|$) to guarantee physical polarization states.
- Implemented a consensus ADMM solver in PyTorch with L_1 spectral sparsity and Sherman–Morrison linear solves; runs on GPU via SLURM.
- Calibrated PSF libraries using Gram–Schmidt orthogonalization to remove channel cross-talk; spectral unmixing via non-negative least squares.

Research Assistant

UVA (Prof. Liqin Ke’s Group)

Oct 2025 – Present

Charlottesville, VA

- Used Density Functional Theory (DFT) to study transition-metal magnetic anisotropy and its relationship to structural motifs in crystal lattices.
- Built interactive 3D atomic-structure visualizations (Three.js, JSmol) for real-time structural inspection.
- Developed a shared ML pipeline (Go, Next.js) for collaborators to run predictive models on magnetic-property datasets.
- Designing RareEarth DB, a rare-earth analogue of the Materials Project: 4,648 curated anisotropy measurements linked to ICSD/Pearson structures, with Fireworks-orchestrated DFT workflows on UVA HPC; DOE funding proposal in preparation.

Research Assistant

UVA BAL Lab (Prof. Xin Cynthia Tong’s Group)

Apr 2026 – Present

Charlottesville, VA

- Developed PRISM, a constrained optimization framework for multivariate longitudinal missing-data imputation: projects initial imputations onto the model-implied covariance structure to correct the variance attenuation of standard conditional-expectation methods.
- Built the solver in C++17 (RcppArmadillo): projected gradient descent with Armijo line search, exact mean-constraint projection, and KKT convergence checks.

Research Intern

UVA Biocomplexity Institute

May 2026 – Aug 2026

Charlottesville, VA

- Engineered a Julia simulation engine for Independent Cascade diffusion and trained PyTorch Geometric GCNs to estimate total epidemic cascade size from sparse, partially observed network infections.
- Developed an agentic workflow engine (Pydantic AI, NetworkX) translating natural-language queries into verified execution DAGs for network-analysis pipelines, with deterministic validation and graph deduplication cutting redundant computation by $> 40\%$.

OPEN SOURCE SOFTWARE

- **stim**: Authored PR #1099 to the quantum stabilizer circuit simulator (Google; accepted in review).
- **PRISM**: C++17 constrained-optimization engine for longitudinal missing-data imputation (see research above).
- **qudec**: Simulation toolkit for quantum LDPC decoding (BP+OSD, LP, ADMM); characterized BP failure modes on degenerate codes and benchmarked decoders under phenomenological noise.

- **sanjaya**: Academic literature extraction pipeline (OpenAlex API, Scrapy, Playwright) for structured bilingual dataset export.
- **difi**: Go-based Git diff TUI; Homebrew Core, 365+ GitHub stars, #5 on Hacker News.

HONORS AND AWARDS

- **Double Hoo Research Grant**, University of Virginia, 2026
- **Dean's List of Distinguished Students**, University of Virginia
- **Silver Division**, USA Computing Olympiad (USACO)

TECHNICAL SKILLS

- **Math & Physics**: Advanced Theoretical Quantum Mechanics, Quantum Information Theory, Convex Optimization, Matrix Calculus, Linear Algebra.
- **Machine Learning & Computing**: PyTorch/CUDA, JAX, Scikit-Learn, Pandas, RcppArmadillo.
- **Languages**: Python, C/C++, Julia, Rust, Go, TypeScript, R, SQL, Bash, $\text{\LaTeX}$.
- **Infrastructure & Cloud**: Linux (openSUSE/RHEL), Docker, K3s Kubernetes, Slurm HPC, Git.

COMMUNITY SERVICE

Volunteer Firefighter

Plainfield Fire Department

2022 – 2024

Plainfield

- Responded to over 200 emergency medical, fire, and rescue calls annually, operating effectively in high-stress environments to provide critical incident mitigation and community support.